# Algebraic tunnelling for the restricted fractional Laplacian on distant domains

August 2026

**Abstract.** Let $D \subset \mathbb{R}^d$ be a bounded centrally symmetric Lipschitz domain, and let $\Omega_L$ be the union of two reflected translates of $D$ whose centers are separated by $L$. For the restricted fractional Laplacian $(-\Delta)^s$, $0 < s < 1$, we determine the leading splitting of the two eigenvalues converging to the one-well ground-state energy. If $\mu_1$ is the one-well ground-state eigenvalue, $\phi_1$ is its positive $L^2$-normalized eigenfunction, and $m_1 = \int_D \phi_1$, then

$$\lambda_1(\Omega_L) = \mu_1 - c_{d,s} m_1^2 L^{-d-2s} + O\Big(L^{-d-2s-2} + L^{-2d-4s}\Big),$$

$$\lambda_2(\Omega_L) = \mu_1 + c_{d,s} m_1^2 L^{-d-2s} + O\Big(L^{-d-2s-2} + L^{-2d-4s}\Big).$$

The proof uses an exact reflection reduction to the two operators $A_D \pm B_L$, an elementary isolated-eigenvalue estimate, and a second-order Taylor expansion of the interaction kernel. For two equal balls, the upper formula gives the leading rate in the nonlocal Hong–Krahn–Szegő minimizing sequence. More generally, for any fixed collection of distinct centers $a_1, \ldots, a_N$, the first $N$ eigenvalues are

$$\lambda_k = \mu_1 + L^{-d-2s}\theta_k + O\Big(L^{-d-2s-2} + L^{-2d-4s}\Big), \qquad 1 \le k \le N,$$

where $\theta_1 \le \cdots \le \theta_N$ are the eigenvalues of the explicit matrix with zero diagonal and off-diagonal entries $-c_{d,s} m_1^2 |a_i - a_j|^{-d-2s}$. Exact cell-integral Galerkin computations on two and three intervals reproduce these limits.

**Keywords.** restricted fractional Laplacian; eigenvalue splitting; nonlocal double wells; spectral perturbation; shape optimization.

**MSC2020 Classification.** 35P15; 35R11.

## 1. Introduction

The restricted fractional Laplacian couples disjoint components through its integral kernel. This differs from the local Dirichlet Laplacian, whose spectrum on a disconnected union is exactly the multiset union of the component spectra. Two distant congruent components therefore form a nonlocal analogue of a double well: their limiting degeneracies split at an algebraic, rather than exponential, scale.

Brasco and Parini proved that the fractional Hong–Krahn–Szegő bound for the second eigenvalue is strict and that two equal balls form a minimizing sequence when their mutual distance tends to infinity [BP16, Theorem 6.2]. Parini and Salort placed such distant components in a general concentration–compactness dichotomy [PS20, Theorems 1.2 and 1.3].

Those results identify the limit but do not specify the leading interaction coefficient. The purpose of this article is to extract that coefficient and the associated effective finite-dimensional interaction.

The influence of mutual position on the restricted fractional spectrum is also emphasized by Abatangelo, Felli, and Noris [AFN20, Introduction, p. 3], in a study of small fractional-capacity perturbations. In the shape-optimization setting, Zahl isolated the exact cross-energy under translations and proposed a signed point-mass interaction with kernel $|x-y|^{-d-2s}$ [Zah26, Section 10, equations (10.3) and (10.7)]. That discrete energy anticipates the matrix in (4.9); the result here derives it from the full restricted fractional operator, bounds the off-diagonal block in operator norm, controls its coupling to the complementary one-well modes through the spectral gap, and identifies every ordered eigenvalue in the cluster. We are not aware of an earlier result giving either the two-component coefficient with a quantified remainder or this finite-well spectral reduction.

The first result, Theorem 3.3, treats the double ground-state cluster. Theorem 4.6 identifies the entire ground-state cluster for finitely many translated components with the spectrum of an explicit weighted interaction matrix. Both proofs are self-contained after the standard one-well spectral facts. The interval provides an explicit benchmark: one-well eigenvalue and eigenfunction estimates are available in [Kwa12, Theorem 1 and Propositions 1–2]; sharper eigenvalue asymptotics and an eigenfunction bound uniform in both the index and the fractional order are proved in [Zha26, Theorems 1 and 2].

## 2. The one-well operator and exact two-well reduction

Fix $d \geq 1$ and $0 < s < 1$, and put

$$\kappa = d + 2s.$$

For an open set $G \subset \mathbb{R}^d$, let

$$\widetilde{H}^s(G) = \{u \in H^s(\mathbb{R}^d) : u = 0 \text{ a.e. on } \mathbb{R}^d \setminus G\}.$$

The zero-exterior quadratic form is

$$\mathrm{II}_G[u] = \frac{c_{d,s}}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x)-u(y)|^2}{|x-y|^\kappa}\, dx\, dy, \qquad u \in \widetilde{H}^s(G), \tag{2.1}$$

where $c_{d,s} > 0$ is the normalization in the singular-integral definition of $(-\Delta)^s$. We denote the associated self-adjoint operator on $L^2(G)$ by $A_G$.

Let $D \subset B_R(0)$ be a bounded Lipschitz domain satisfying $D = -D$. The Lipschitz extension property and compactness of the form embedding follow from [DNPV12, Theorems 5.4 and 7.1]. Write the first two eigenvalues of $A_D$ as

$$\mu_1 < \mu_2, \qquad g = \mu_2 - \mu_1 > 0.$$

The first eigenvalue is simple and has a strictly positive normalized eigenfunction by [BP16, Theorem 2.8], specialized to the linear case. Fix that eigenfunction and set

$$A_D\phi_1 = \mu_1\phi_1, \qquad \|\phi_1\|_{L^2(D)} = 1, \qquad m_1 = \int_D \phi_1(x)\, dx > 0. \tag{2.2}$$

Simplicity and reflection invariance imply $\phi_1(-x) = \phi_1(x)$.

Fix a unit vector $e \in \mathbb{R}^d$. For $L > 2R$, define

$$D_L^- = D - \frac{L}{2}e, \qquad D_L^+ = D + \frac{L}{2}e, \qquad \Omega_L = D_L^- \cup D_L^+.$$

On $L^2(D)$ define the bounded integral operator

$$(B_L v)(x) = -c_{d,s} \int_D \frac{v(y)}{|Le - x - y|^\kappa}\, dy. \tag{2.3}$$

Its kernel is real and symmetric, so $B_L$ is self-adjoint.

**Lemma 2.1** *(Exact reflection reduction)*. *The operator $A_{\Omega_L}$ is unitarily equivalent to*

$$\begin{pmatrix} A_D & B_L \\ B_L & A_D \end{pmatrix} \quad \text{on } L^2(D) \oplus L^2(D),$$

*and hence to*

$$(A_D + B_L) \oplus (A_D - B_L). \tag{2.4}$$

*Proof.* For $f \in \widetilde{H}^s(\Omega_L)$ set

$$u(x) = f\left(x - \frac{L}{2}e\right), \qquad v(x) = f\left(-x + \frac{L}{2}e\right), \qquad x \in D.$$

This defines a unitary map from $L^2(\Omega_L)$ to $L^2(D) \oplus L^2(D)$. Translational and reflection invariance identify the two diagonal parts of the form with $\mathrm{II}_D[u]$ and $\mathrm{II}_D[v]$. Because the two components have positive separation, their cross-kernel is bounded; consequently this unitary maps $\widetilde{H}^s(\Omega_L)$ onto $\widetilde{H}^s(D) \oplus \widetilde{H}^s(D)$. Expanding the square in (2.1), the difference between the form of the sum and the two separate forms is exactly

$$-2c_{d,s}\,\mathrm{Re} \iint_{D\times D} \frac{u(x)\overline{v(y)}}{|Le - x - y|^\kappa}\, dx\, dy = 2\,\mathrm{Re}\langle u, B_L v\rangle.$$

This proves the block representation in the sense of closed forms. Since $B_L$ is bounded, it also gives the stated operator representation. The unitary change of variables $(u, v) \mapsto 2^{-1/2}(u + v, u - v)$ yields (2.4). ∎

## 3. The doublet splitting

We first record the elementary perturbation estimate used for the two scalar branches.

**Lemma 3.1** *(Ground-state perturbation bound)*. *Let $A$ be self-adjoint with compact resolvent, simple lowest eigenvalue $\mu$, normalized ground state $\phi$, and spectral gap $g > 0$. Let $W$ be bounded and self-adjoint, put $b = \|W\|$ and $w = \langle \phi, W\phi\rangle$, and suppose $b \le g/4$. Then*

$$-\frac{2b^2}{g} \le \lambda_1(A + W) - (\mu + w) \le 0. \tag{3.1}$$

*Proof.* The upper bound follows by testing the Rayleigh quotient with $\phi$. For a normalized vector $\psi = a\phi + \eta$, where $\eta \perp \phi$ and $t = \|\eta\|$, the spectral gap gives

$$\langle \psi, A\psi \rangle \geq \mu + gt^2.$$

Since $|w| \leq b$ and $|a| \leq 1$,

$$\begin{aligned} \langle \psi, (A+W)\psi \rangle - (\mu + w) &\geq gt^2 - 2bt^2 - 2bt \\ &= (g-2b)t^2 - 2bt \\ &\geq -\frac{b^2}{g-2b} \geq -\frac{2b^2}{g}. \end{aligned}$$

Taking the infimum over normalized $\psi$ proves the lower bound. ∎

The next lemma extracts the interaction coefficient and records why central symmetry improves the kernel error by one power of $L$.

**Lemma 3.2** *(Interaction-kernel expansion)***.** *Let*

$$I_L = \iint_{D\times D} \frac{\phi_1(x)\phi_1(y)}{|Le - x - y|^\kappa}\, dx\, dy.$$

*For $L \geq 4R$,*

$$|I_L - m_1^2 L^{-\kappa}| \leq C_{\kappa,R} m_1^2 L^{-\kappa-2}, \qquad C_{\kappa,R} = 2^{\kappa+3}\kappa(\kappa+3)R^2. \tag{3.2}$$

*Moreover,*

$$\|B_L\| \leq \beta_L, \qquad \beta_L = c_{d,s} 2^\kappa |D|\, L^{-\kappa}. \tag{3.3}$$

*Proof.* Set $F_L(z) = |Le - z|^{-\kappa}$. For $|z| \leq 2R$ and $L \geq 4R$, the segment from 0 to $z$ remains at distance at least $L/2$ from $Le$. Direct differentiation gives

$$\|D^2 F_L(z)\| \leq \kappa(\kappa+3)|Le - z|^{-\kappa-2}.$$

Taylor's theorem therefore yields

$$|F_L(z) - L^{-\kappa} - \kappa L^{-\kappa-1} e\cdot z| \leq C_{\kappa,R} L^{-\kappa-2}$$

for $|z| \leq 2R$, with the displayed value of $C_{\kappa,R}$. Because $\phi_1$ is even,

$$\int_D x\phi_1(x)\, dx = 0.$$

Consequently the integral of the linear Taylor term with $z = x + y$ vanishes. Since $\phi_1$ is positive, integration of the remainder gives (3.2).

Finally, $|Le - x - y| \geq L - 2R \geq L/2$. The Schur test applied to (2.3) gives

$$\|B_L\| \leq c_{d,s}|D|(L-2R)^{-\kappa} \leq c_{d,s} 2^\kappa |D|\, L^{-\kappa},$$

which is (3.3). ∎

**Theorem 3.3** *(Two-well algebraic splitting)***.** *Let $D \subset B_R(0)$ be a bounded centrally symmetric Lipschitz domain, and let $\Omega_L$ be defined above. There is $L_0 < \infty$, determined by $d, s, D, R$ and the one-well gap $g$, such that for $L \geq L_0$ the first two eigenvalues of $A_{\Omega_L}$ satisfy*

$$|\lambda_1(\Omega_L) - (\mu_1 - c_{d,s} m_1^2 L^{-\kappa})| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}, \tag{3.4}$$

$$|\lambda_2(\Omega_L) - (\mu_1 + c_{d,s} m_1^2 L^{-\kappa})| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}. \tag{3.5}$$

*In particular,*

$$L^{d+2s}\big(\lambda_2(\Omega_L) - \lambda_1(\Omega_L)\big) \longrightarrow 2c_{d,s} m_1^2. \tag{3.6}$$

*Proof.* By Lemma 2.1, the spectrum is the union of the spectra of $A_D + B_L$ and $A_D - B_L$. Put

$$t_L = \langle \phi_1, B_L \phi_1 \rangle = -c_{d,s} I_L < 0.$$

Choose $L_0 \geq 4R$ so large that, for $L \geq L_0$,

$$\beta_L \leq \frac{g}{4}, \qquad 2c_{d,s} m_1^2 (L + 2R)^{-\kappa} > \frac{4\beta_L^2}{g}. \tag{3.7}$$

Such a choice exists because the left-hand side of the second inequality is of order $L^{-\kappa}$ and its right-hand side is of order $L^{-2\kappa}$.

Apply Lemma 3.1 first with $W = B_L$ and then with $W = -B_L$. The two branch ground-state energies have the form

$$E_+(L) = \mu_1 + t_L + \rho_+(L), \qquad E_-(L) = \mu_1 - t_L + \rho_-(L), \tag{3.8}$$

where

$$|\rho_\pm(L)| \leq \frac{2\beta_L^2}{g}.$$

The second eigenvalue of either branch is at least $\mu_2 - \beta_L \geq \mu_1 + 3g/4$, whereas both branch ground-state energies are at most $\mu_1 + \beta_L \leq \mu_1 + g/4$. Thus the first two eigenvalues of the full operator are the two numbers in (3.8).

Since $|Le - x - y| \leq L + 2R$ and $\phi_1 > 0$,

$$I_L \geq m_1^2 (L + 2R)^{-\kappa}.$$

The second condition in (3.7) and (3.8) imply $E_+(L) < E_-(L)$. Hence $\lambda_1(\Omega_L) = E_+(L)$ and $\lambda_2(\Omega_L) = E_-(L)$. Equations (3.4) and (3.5) now follow from Lemma 3.2. Subtracting the two expansions and multiplying by $L^\kappa$ proves (3.6), since $\kappa > 0$. ∎

**Corollary 3.4** *(Rate for the distant-ball minimizing sequence)***.** *Let $D = B_R(0)$ and let $\Omega_L$ be the union of two radius-$R$ balls whose centers are separated by $L$. Then*

$$L^{d+2s}\big(\lambda_2(\Omega_L) - \lambda_1(B_R)\big) \longrightarrow c_{d,s} \left( \int_{B_R} \phi_1(x)\, dx \right)^2.$$

*Thus the minimizing sequence in [BP16, Theorem 6.2] approaches its limiting value at the explicit leading order $L^{-d-2s}$.*

*Proof.* This is (3.5) with $D = B_R(0)$ and $\mu_1 = \lambda_1(B_R)$. ■

## 4. Finitely many wells

The two-well reflection decomposition is special to a pair. For finitely many components we instead keep the full block operator and compare its lowest spectral cluster with its compression to the translated one-well ground states.

**Definition 4.1** (Multi-well geometry)**.** Fix an integer $N \geq 2$ and distinct points $\mathbf{a} = (a_1, \ldots, a_N)$ in $\mathbb{R}^d$. Set

$$\delta_{\mathbf{a}} = \min_{i\neq j}|a_i - a_j|, \qquad \Sigma_p(\mathbf{a}) = \max_{1\leq i\leq N} \sum_{j\neq i} |a_i - a_j|^{-p} \quad (p > 0). \tag{4.1}$$

For $L\delta_{\mathbf{a}} > 2R$, define

$$D_{j,L} = D + La_j, \qquad \Omega_{\mathbf{a},L} = \bigcup_{j=1}^{N} D_{j,L}. \tag{4.2}$$

The sets in this union are pairwise disjoint.

On $\mathcal{H}_N = \bigoplus_{j=1}^N L^2(D)$, let $\mathcal{A}_0 = \bigoplus_{j=1}^N A_D$. For $i \neq j$ define

$$(B_{ij,L}v)(x) = -c_{d,s}\int_D \frac{v(y)}{|L(a_i - a_j) + x - y|^\kappa}\, dy, \qquad B_{ii,L} = 0, \tag{4.3}$$

and let $\mathcal{V}_L = (B_{ij,L})_{i,j=1}^N$.

**Lemma 4.2** *(Exact multi-well reduction and norm bound)***.** *If $L\delta_{\mathbf{a}} \geq 4R$, then $A_{\Omega_{\mathbf{a},L}}$ is unitarily equivalent to $\mathcal{A}_0 + \mathcal{V}_L$. Moreover,*

$$\|\mathcal{V}_L\| \leq \gamma_L, \qquad \gamma_L = c_{d,s}2^\kappa |D|\, \Sigma_\kappa(\mathbf{a}) L^{-\kappa}. \tag{4.4}$$

*Proof.* For $f \in L^2(\Omega_{\mathbf{a},L})$, put $u_j(x) = f(x + La_j)$ for $x \in D$. This is a unitary identification with $\mathcal{H}_N$. Translation invariance gives the diagonal operator $\mathcal{A}_0$. Positive separation makes every cross-kernel bounded, so the form domain is identified with $\bigoplus_{j=1}^N \widetilde{H}^s(D)$. For $i < j$, direct expansion of the form (2.1) gives the cross term

$$-2c_{d,s}\,\mathrm{Re} \iint_{D\times D} \frac{u_i(x)\overline{u_j(y)}}{|L(a_i - a_j) + x - y|^\kappa}\, dx\, dy.$$

Since $B_{ji,L} = B_{ij,L}^*$, summing these terms proves the exact block representation in the sense of closed forms and hence as operators.

For $i \neq j$ and $x, y \in D$,

$$|L(a_i - a_j) + x - y| \geq L|a_i - a_j| - 2R \geq \frac{L}{2}|a_i - a_j|.$$

The Schur test therefore gives

$$\|B_{ij,L}\| \leq c_{d,s}2^\kappa |D|\, L^{-\kappa}|a_i - a_j|^{-\kappa}. \tag{4.5}$$

For $u = (u_1, \ldots, u_N)$, set $r_j = \|u_j\|_{L^2(D)}$ and let $G$ be the symmetric matrix with entries $G_{ij} = \|B_{ij,L}\|$. Then

$$\|\mathcal{V}_L u\|_{\mathcal{H}_N} \le \|Gr\|_{\ell^2} \le \left(\max_i \sum_j G_{ij}\right) \|r\|_{\ell^2}.$$

Combining this inequality with (4.5) proves (4.4). ■

We next record the finite-multiplicity perturbation estimate that replaces Lemma 3.1.

**Lemma 4.3** *(Isolated spectral-cluster bound)***.** *Let $A$ be self-adjoint with compact resolvent. Suppose its lowest eigenvalue $\mu$ has multiplicity $N$, let $P$ be the corresponding spectral projection, and suppose*

$$A\big|_{P^\perp} \ge \mu + g$$

*in the quadratic-form sense for some $g > 0$. Let $W$ be bounded and self-adjoint, set $b = \|W\|$, and assume $b \le g/4$. If $\eta_1 \le \cdots \le \eta_N$ are the eigenvalues of $PWP$ on $\operatorname{ran} P$, then, for $1 \le k \le N$,*

$$-\frac{2b^2}{g} \le \lambda_k(A+W) - (\mu + \eta_k) \le 0. \tag{4.6}$$

*In addition,*

$$\lambda_{N+1}(A+W) \ge \mu + g - b \ge \mu + \frac{3g}{4}. \tag{4.7}$$

*Proof.* The min–max principle applied to the span of the first $k$ eigenvectors of $PWP$ gives the upper bound in (4.6). For the lower bound, write $\psi = p + q$ with $p = P\psi$ and $q = (I-P)\psi$. Then

$$\begin{aligned}\langle \psi, (A+W)\psi\rangle &\ge \mu\|\psi\|^2 + \langle p, PWPp\rangle + (g-b)\|q\|^2 - 2b\|p\|\|q\| \\ &\ge \mu\|\psi\|^2 + \langle p, PWPp\rangle - \frac{2b^2}{g}\|p\|^2 + \left(\frac{g}{2} - b\right)\|q\|^2.\end{aligned}$$

The second line uses $2b\|p\|\|q\| \le 2b^2 g^{-1}\|p\|^2 + (g/2)\|q\|^2$. Thus, in the form sense, $A+W$ dominates the direct sum of

$$\mu I + PWP - \frac{2b^2}{g} I \quad \text{on } \operatorname{ran} P$$

and $(\mu + g/2 - b)I$ on $P^\perp$. Since $\eta_N \le b \le g/4$ and $g/2 - b \ge g/4$, the first $N$ eigenvalues of this comparison operator are the $N$ eigenvalues of its $P$-block. Another application of the min–max principle gives the lower bound in (4.6). Finally, $A + W \ge A - bI$ and the min–max principle give (4.7). ■

Let $P_1$ be the orthogonal projection in $\mathcal{H}_N$ onto the span of the $N$ vectors whose $j$th component is $\phi_1$ and whose other components vanish. In this basis the compression $P_1\mathcal{V}_L P_1$ is represented by the real symmetric matrix $T_L = (t_{ij}(L))$, where

$$t_{ii}(L) = 0, \qquad t_{ij}(L) = -c_{d,s} \iint_{D\times D} \frac{\phi_1(x)\phi_1(y)}{|L(a_i - a_j) + x - y|^\kappa}\, dx\, dy \quad (i \ne j). \tag{4.8}$$

**Definition 4.4** (Effective interaction matrix)**.** Define the real symmetric $N \times N$ matrix $M_{\mathbf{a}}$ by

$$(M_{\mathbf{a}})_{ii} = 0, \qquad (M_{\mathbf{a}})_{ij} = -c_{d,s} m_1^2 |a_i - a_j|^{-\kappa} \quad (i \neq j). \tag{4.9}$$

Write its ordered eigenvalues as $\theta_1 \leq \cdots \leq \theta_N$.

**Lemma 4.5** *(Compression error)***.** *If $L\delta_{\mathbf{a}} \geq 4R$, then*

$$\|T_L - L^{-\kappa} M_{\mathbf{a}}\|_{\ell^2 \to \ell^2} \leq \varepsilon_L, \qquad \varepsilon_L = c_{d,s} C_{\kappa,R} m_1^2 \Sigma_{\kappa+2}(\mathbf{a}) L^{-\kappa-2}. \tag{4.10}$$

*Proof.* Fix $i \neq j$, put $a = a_i - a_j$, and set $F(z) = |La + z|^{-\kappa}$. For $|z| \leq 2R$, the segment from $0$ to $z$ stays at distance at least $L|a|/2$ from $-La$. As in the proof of Lemma 3.2, Taylor's theorem gives

$$|F(z) - F(0) - DF(0)z| \leq C_{\kappa,R} L^{-\kappa-2} |a|^{-\kappa-2}. \tag{4.11}$$

The linear term vanishes after integration because

$$\iint_{D\times D} \phi_1(x)\phi_1(y)(x-y)\,dx\,dy = 0.$$

Using $F(0) = L^{-\kappa}|a|^{-\kappa}$ in (4.8) and integrating (4.11) gives

$$|t_{ij}(L) - L^{-\kappa}(M_{\mathbf{a}})_{ij}| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} |a_i - a_j|^{-\kappa-2}.$$

The operator norm of a real symmetric matrix is at most its largest absolute row sum. Taking that row sum proves (4.10). ∎

**Theorem 4.6** *(Finite multi-well effective Hamiltonian)***.** *Let $D$, $\mu_1$, $g$, $\phi_1$, and $m_1$ be as in Section 2, and let the multi-well geometry be as in Definition 4.1. If*

$$L\delta_{\mathbf{a}} \geq 4R, \qquad \gamma_L \leq \frac{g}{4}, \tag{4.12}$$

*then, for $1 \leq k \leq N$,*

$$|\lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + L^{-\kappa}\theta_k)| \leq \varepsilon_L + \frac{2\gamma_L^2}{g}. \tag{4.13}$$

*The cluster is separated from the rest of the spectrum by*

$$\lambda_{N+1}(\Omega_{\mathbf{a},L}) \geq \mu_1 + g - \gamma_L \geq \mu_1 + \frac{3g}{4}. \tag{4.14}$$

*Consequently, for every $1 \leq k \leq N$,*

$$L^{d+2s}\big(\lambda_k(\Omega_{\mathbf{a},L}) - \mu_1\big) \longrightarrow \theta_k. \tag{4.15}$$

*Proof.* Lemma 4.2 identifies the full operator with $\mathcal{A}_0 + \mathcal{V}_L$. The lowest eigenvalue of $\mathcal{A}_0$ is $\mu_1$, with multiplicity $N$, its spectral projection is $P_1$, and the rest of its spectrum lies in $[\mu_1 + g, \infty)$. Apply Lemma 4.3 with $A = \mathcal{A}_0$, $W = \mathcal{V}_L$, and $b = \|\mathcal{V}_L\| \le \gamma_L$. If $\tau_1(L) \le \cdots \le \tau_N(L)$ are the eigenvalues of $T_L$, this gives

$$-\frac{2\gamma_L^2}{g} \le \lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + \tau_k(L)) \le 0. \tag{4.16}$$

The finite-dimensional min–max principle and Lemma 4.5 give

$$|\tau_k(L) - L^{-\kappa}\theta_k| \le \varepsilon_L.$$

Combining these two estimates proves (4.13), and (4.14) follows from (4.7). Finally, $L^\kappa \varepsilon_L = O(L^{-2})$ and $L^\kappa \gamma_L^2 = O(L^{-\kappa})$; since $\kappa = d + 2s > 0$, multiplying (4.13) by $L^\kappa$ proves (4.15). ■

**Corollary 4.7** *(Recovery of the doublet)***.** *For $N = 2$, $a_1 = -e/2$, and $a_2 = e/2$, the eigenvalues of $M_\mathbf{a}$ are $-c_{d,s}m_1^2$ and $c_{d,s}m_1^2$. The bound (4.13) then agrees with (3.4)–(3.5).*

*Proof.* In this case

$$M_\mathbf{a} = \begin{pmatrix} 0 & -c_{d,s}m_1^2 \\ -c_{d,s}m_1^2 & 0 \end{pmatrix},$$

whose ordered eigenvalues are the two stated numbers. Moreover, $\Sigma_\kappa(\mathbf{a}) = \Sigma_{\kappa+2}(\mathbf{a}) = 1$, so $\gamma_L = \beta_L$ and the two displayed remainder bounds coincide. ■

**Corollary 4.8** *(Collective ground state)***.** *For every multi-well configuration, $\theta_1 < 0$ is simple and has an eigenvector with strictly positive entries. Moreover, $\theta_N > 0$. Consequently, for all sufficiently large $L$, the lowest eigenvalue in the cluster is simple and lies below $\mu_1$, while at least one cluster eigenvalue lies above $\mu_1$.*

*Proof.* The off-diagonal entries of $M_\mathbf{a}$ are strictly negative. Its Rayleigh quotient at $(1, \ldots, 1)$ is negative, so $\theta_1 < 0$. If $z$ is a minimizer of the Rayleigh quotient, replacing $z$ by its componentwise absolute value cannot increase that quotient, and equality forces all nonzero components to have the same sign. The eigenvalue equation then shows that no component can vanish. Thus every eigenvector for $\theta_1$ has one strict sign. Two linearly independent such eigenvectors would have a nonzero linear combination with a zero component, so $\theta_1$ is simple. Finally, $\operatorname{tr} M_\mathbf{a} = 0$ and $\theta_1 < 0$, hence $\theta_N > 0$. The assertions for the full operator follow from (4.13) and the strict gaps between the relevant eigenvalues of $M_\mathbf{a}$. ■

**Corollary 4.9** *(Regular-simplex cluster)***.** *Suppose the centers satisfy $|a_i - a_j| = r$ for all $i \ne j$, as for the vertices of a regular simplex. Put*

$$w = c_{d,s}m_1^2 r^{-\kappa}.$$

*Then $\theta_1 = -(N-1)w$ and $\theta_2 = \cdots = \theta_N = w$. Thus one collective level is lowered by $(N-1)wL^{-\kappa}$, while the remaining $N-1$ levels are raised by $wL^{-\kappa}$, up to the remainder in (4.13).*

*Proof.* Here $M_\mathbf{a} = -w(J - I)$, where $J$ is the all-ones matrix. The vector $(1, \ldots, 1)$ has eigenvalue $-(N-1)w$, and its orthogonal complement has eigenvalue $w$. ■

## 5. A reproducible interval check

We finish with a conforming Galerkin check of the two- and three-well limits. It is included as a reproducibility test of the formulas, not as an input to their proofs. Take $d = 1$, $0 < s < 1/2$, and partition each unit interval into cells of length $h$. Proposition 5.1 shows directly that cell indicators have finite form energy in this range, so they span a conforming finite-dimensional subspace of the form domain.

**Proposition 5.1** *(Exact cell-integral matrix)***.** *Let $I_\alpha$ be pairwise disjoint cells of length $h$, let $x_\alpha$ be their centers, put $r_{\alpha\beta} = |x_\alpha - x_\beta|$, and set $q = 1 - 2s$. In the $L^2$-orthonormal basis $e_\alpha = h^{-1/2}\mathbf{1}_{I_\alpha}$, the Galerkin matrix $H_h$ of the zero-exterior form has entries*

$$(H_h)_{\alpha\alpha} = \frac{c_{1,s}}{h}\frac{2h^q}{2sq}, \tag{5.1}$$

$$(H_h)_{\alpha\beta} = -\frac{c_{1,s}}{h}\frac{2r^q_{\alpha\beta} - (r_{\alpha\beta} + h)^q - (r_{\alpha\beta} - h)^q}{2sq}, \qquad \alpha \neq \beta. \tag{5.2}$$

*Proof.* For two disjoint cells with center distance $r \geq h$, direct integration twice gives

$$\iint_{I_\alpha \times I_\beta} |x - y|^{-1-2s}\, dx\, dy = \frac{2r^q - (r + h)^q - (r - h)^q}{2sq}.$$

The off-diagonal form entry is minus $c_{1,s}$ times this integral. For one cell,

$$\int_{I_\alpha}\int_{\mathbb{R}\setminus I_\alpha} |x - y|^{-1-2s}\, dy\, dx = \frac{2h^q}{2sq}.$$

Dividing both expressions by $h$ for the normalized indicators proves (5.1)–(5.2). ■

**Corollary 5.2** *(Fixed-mesh cluster asymptotic)***.** *Fix a uniform partition of $D = (-1/2, 1/2)$ and let $\mu_{1,h}$ and $\phi_{1,h}$ be the lowest eigenvalue and a positive, $L^2$-normalized piecewise-constant eigenfunction of its Galerkin matrix. Put $m_{1,h} = \int_D \phi_{1,h}$. For the corresponding Galerkin matrix on $\bigcup_{j=1}^N (D + La_j)$, write its ordered eigenvalues as $\lambda_{k,h}(L)$. Then, for fixed distinct $a_1, \dots, a_N$,*

$$\lambda_{k,h}(L) = \mu_{1,h} + L^{-1-2s}\theta_{k,h} + O_{h,\mathbf{a},s}\big(L^{-3-2s} + L^{-2-4s}\big), \qquad 1 \leq k \leq N, \tag{5.3}$$

*where $\theta_{1,h} \leq \cdots \leq \theta_{N,h}$ are the eigenvalues of the matrix with zero diagonal and off-diagonal entries $-c_{1,s}m^2_{1,h}|a_i - a_j|^{-1-2s}$.*

*Proof.* All off-diagonal entries of the one-well matrix in (5.2) are negative. The Rayleigh-quotient argument in the proof of Corollary 4.8 therefore gives a simple lowest eigenvalue and a strictly positive eigenvector. On the direct sum of $N$ fixed Galerkin spaces, the block expansion used in Lemma 4.2 is exact, its off-diagonal norm is $O_{h,\mathbf{a},s}(L^{-1-2s})$, and the complement of the $N$ copied ground states is separated by the positive discrete one-well gap. The Taylor argument in Lemma 4.5, now in Euclidean operator norm, gives an $O_{h,\mathbf{a},s}(L^{-3-2s})$ compression error. Applying Lemma 4.3 to this finite matrix gives the additional $O_{h,\mathbf{a},s}(L^{-2-4s})$ term and proves (5.3). ■

For the computation we take $s = 1/4$, $D = (-1/2, 1/2)$, and the standard normalization

$$c_{1,s} = \frac{4^s s\,\Gamma(1/2+s)}{\sqrt{\pi}\,\Gamma(1-s)}.$$

Table 1 records the one-well quantities. The stabilization of $m_{1,h}$ is the relevant check for the interaction coefficient.

| cells | $\mu_{1,h}$ | $\mu_{2,h} - \mu_{1,h}$ | $m_{1,h}$ |
|---|---|---|---|
| 60 | 1.3794549573 | 0.9105539931 | 0.9628386433 |
| 120 | 1.3753692130 | 0.8998883318 | 0.9633505837 |
| 180 | 1.3741292905 | 0.8969647744 | 0.9636021778 |

**Table 1.** One-well Galerkin data for $s = 1/4$.

On the 180-cell mesh, the predicted two-well scaled levels are $(-0.18521477, 0.18521477)$. For three collinear centers $(-L, 0, L)$, they are $(-0.29671332, 0.06548331, 0.23123001)$. The direct Galerkin eigenvalues are shown in Table 2; the last column is the largest absolute difference from the corresponding effective-matrix eigenvalue.

| $L$ | $L^{3/2}(\lambda_{1,h} - \mu_{1,h})$ | $L^{3/2}(\lambda_{2,h} - \mu_{1,h})$ | max. error |
|---|---|---|---|
| 3 | $-0.190740$ | 0.189832 | $5.52 \times 10^{-3}$ |
| 6 | $-0.186544$ | 0.186335 | $1.33 \times 10^{-3}$ |
| 12 | $-0.185551$ | 0.185486 | $3.36 \times 10^{-4}$ |
| 24 | $-0.185286$ | 0.185295 | $8.04 \times 10^{-5}$ |

| $L$ | $L^{3/2}(\lambda_{1,h} - \mu_{1,h})$ | $L^{3/2}(\lambda_{2,h} - \mu_{1,h})$ | $L^{3/2}(\lambda_{3,h} - \mu_{1,h})$ | max. error |
|---|---|---|---|---|
| 3 | $-0.304984$ | 0.065562 | 0.237544 | $8.27 \times 10^{-3}$ |
| 6 | $-0.298738$ | 0.065554 | 0.232747 | $2.03 \times 10^{-3}$ |
| 12 | $-0.297232$ | 0.065510 | 0.231597 | $5.19 \times 10^{-4}$ |
| 24 | $-0.296843$ | 0.065504 | 0.231323 | $1.29 \times 10^{-4}$ |

**Table 2.** Scaled two-well (top) and three-well (bottom) cluster shifts.

The scaled errors decrease to zero in both configurations, as asserted at fixed mesh by Corollary 5.2. The calculation uses the exact entries (5.1)–(5.2); no singular quadrature or fitted coefficient enters the tables.

## Code availability

The script `tools/check_asymptotics.py` reconstructs every entry in Tables 1 and 2. It checks the cell-integral identity before solving the eigenproblems and prints progress after each separation. The pinned numerical dependencies are listed in `requirements-numerics.txt`.

## References


[AFN20] Laura Abatangelo, Veronica Felli, and Benedetta Noris. On simple eigenvalues of the fractional Laplacian under removal of small fractional capacity sets. *Communications in Contemporary Mathematics*, 22(8):1950071, 2020.

[BP16] Lorenzo Brasco and Enea Parini. The second eigenvalue of the fractional $p$-laplacian. *Advances in Calculus of Variations*, 9(4):323–355, 2016.

[DNPV12] Eleonora Di Nezza, Giampiero Palatucci, and Enrico Valdinoci. Hitchhiker's guide to the fractional Sobolev spaces. *Bulletin des Sciences Mathématiques*, 136(5):521–573, 2012.

[Kwa12] Mateusz Kwaśnicki. Eigenvalues of the fractional Laplace operator in the interval. *Journal of Functional Analysis*, 262(5):2379–2402, 2012.

[PS20] Enea Parini and Ariel Salort. Compactness and dichotomy in nonlocal shape optimization. *Mathematische Nachrichten*, 293(11):2208–2232, 2020.

[Zah26] Alvis Zahl. Minimizing eigenvalues of the fractional Laplacian. *Calculus of Variations and Partial Differential Equations*, 65:98, 2026.

[Zha26] Cheng Zhang. Eigenvalue asymptotics and uniform eigenfunction bounds for the fractional Laplacian in the interval, 2026. arXiv:2608.23457 [math.CA].